Homework 13

For Thursday, November 5, 2026.

Give exact answers and justify your reasoning. Do not use electronic tools. All functions are real-valued, angles are measured in radians, and all integrals are Riemann integrals.

1. Compute

$$\int_0^2 (1+x^2)\,dx$$

using right-endpoint Riemann sums with n equal subintervals, without using an antiderivative. Explain why these sums tend to the integral. You may use $\sum_{i=1}^{n} i^2 = n(n+1)(2n+1)/6$.

2. Let f, g be continuous on $[-1, 3]$, with

$$\int_{-1}^{1} f(x)\,dx = 2, \qquad \int_{1}^{3} f(x)\,dx = -1, \qquad \int_{-1}^{3} g(x)\,dx = 4.$$

Determine $\displaystyle\int_3^{-1} (2f(x) - 3g(x) + 1)\,dx$.

3. Without finding an antiderivative, use comparison to prove that

$$\frac{2}{3} \le \int_0^1 \frac{1}{1+x^2}\,dx \le 1.$$

4. Is the identity

$$\left|\int_{-1}^{1} f(x)\,dx\right| = \int_{-1}^{1} |f(x)|\,dx$$

true for every continuous function f on $[-1, 1]$? Prove it or give a counterexample, calculating both sides.

5. Evaluate

$$\int_1^4 \left(\sqrt{x} + \frac{1}{x}\right)\,dx$$

using the Fundamental Theorem of Calculus. State an antiderivative and verify its derivative on the interval.

6. Use the product rule to find a function F with $F'(x) = \sin x + x\cos x$. Verify the derivative, and then evaluate

$$\int_0^{\pi/2} (\sin x + x\cos x)\,dx.$$

7. Define F on $[0,1]$ by

$$F(x) = \begin{cases} x, & 0 \le x < 1, \\ 2, & x = 1. \end{cases}$$

Although $F'(x) = 1$ for $0 < x < 1$, compare $\int_0^1 1\,dx$ with $F(1) - F(0)$. Which hypothesis of the evaluation part of the Fundamental Theorem fails?

8. For $-1 \le x \le 1$, let

$$A(x) = \int_{-1}^{x} |t|\,dt.$$

Find a piecewise formula for A. Then verify $A'(0) = 0$ directly from the definition of the derivative.

9. Find the domain and derivative of

$$H(x) = \int_0^{\ln x} \sqrt{1 + t^4}\,dt$$

without evaluating the integral.

10. Use the Fundamental Theorem of Calculus to find

$$\lim_{h \to 0} \frac{1}{h} \int_2^{2+h} \sqrt{1 + t^3}\,dt.$$

11. Find all real numbers x satisfying

$$\int_0^x (t^2+1)\,dt = \frac{4}{3}.$$

12. Find all pairs consisting of a continuous function $f : \mathbb{R} \to \mathbb{R}$ and a constant $c \in \mathbb{R}$ such that

$$\int_c^x f(t)\,dt = x^2 - 4 \qquad \text{for every } x \in \mathbb{R}.$$

Verify that every pair you find satisfies the identity.

13. Let f be continuous on $[0,1]$, and suppose $\int_0^1 f(x)\,dx = 2$. Prove that

$$\int_0^1 f(x)^2\,dx \geq 4,$$

and give a function for which equality holds. You may use $(f(x)-2)^2 \geq 0$.

14. Let f be bounded on $[a,b]$, where $a < b$. Suppose I and J both satisfy the epsilon–delta definition of the Riemann integral of f on $[a,b]$. Prove directly from that definition that $I = J$.

15. Let f be continuous on $[a,b]$, with $a < b$, and suppose

$$f(x) \geq 0 \quad \text{for every } x \in [a,b], \qquad \int_a^b f(x)\,dx = 0.$$

Prove that $f(x) = 0$ for every $x \in [a,b]$. If $f(c) > 0$, use continuity to obtain a positive lower bound on a subinterval of positive length, and then use comparison.